Sweatshop on P-Chess: Review exploration

Side lines in the Review (REV-007), engine lines and live Stockfish (REV-008) and live classes for side-line moves (REV-009); GPT-6.1 Terra lineup, lane `review-explore`

Foreman, session `sweatshop/review-explore/2026-11-05-1723`

5 November 2026

Summary

First run of the `review-explore` lane, in a new clone at `.artifacts-build/lanes/review-explore` made from `main` at `184a0bf` (the spec and tickets from Bruno's grilling the same day). Three launches: 17:23 (one stage, then nothing runnable), 17:39 after the environmental fix, 18:21 after the reopen diagnosis. Last merge at 20:04, session PR #42.

- **Produced:** the lane's three tickets merged on the session with `Verdict: Approve`: REV-008 and REV-009 (`Review: human`: Bruno checks them with real Stockfish on Windows before merging into `main`) and REV-007 (`Review: agent`). One follow-up filed by review: CLEAN-014 (needs-triage).
- **Cost: $11.40** at API list price over 15 stages, all priced: `gpt-6.1-terra xhigh` implementing $1.41 (3), `gpt-6.1-sol xhigh` implementing hard tickets $4.57 (5), `gpt-6.1-sol xhigh` reviewing $5.43 (7). Codex quota: 8 points of the weekly window (Sol xhigh 4: 2 implementing, 2 reviewing; Terra xhigh 0; 4 outside the stages, mostly the EasyMath `infra` lane running at the same time and one foreman `codex exec` check; ±1 per stage, the total is exact), no reset. Plan: the account's Codex plan, shared with the EasyMath lane.
- **What went wrong:** the first stage never loaded `ticket-flow`: the skills were not installed for Codex on the Linux box (no `~/.agents/skills`) and P-Chess's `AGENTS.md` does not tell a bare-id prompt to load it. The stage coded REV-007 straight on the session, with no branch, commits or test-first, and the driver parked it. Then REV-007 took three reopens, the last two on narrower variants of the same Settings-access criterion.
- **Provisional verdict:** the feature is built and reviewed with real viewport input and fake engines. Its central risk, live Stockfish, is unexamined: no engine runs on this machine. Bruno's Windows check is the gate to `main`.

What was done

REV-007: in the Review you can now play your own move from any position of the game (drag or click-click). It starts one temporary side line, listed as extra rows under the branching move with a **Back to game** button. Arrows and Previous/Next walk the line; clicking a game move, the graph, First/Last, Back to game or Esc leaves it, and nothing of it is saved or exported. A new *Best move arrow* setting (default on) draws the stored best move in translucent blue on game positions, and the Review's board now follows the board preferences live. Main files `ui/review.gd`, `ui/main.gd`, `ui/board_view.gd`, `core/settings.gd`. Reopened three times: (1) the Review board ignored real mouse input: `ui/review.tscn` left it at `MOUSE_FILTER_IGNORE`, and the tests called `board.play()` directly. Settings could not open over the Review: `_covers_settings()` in `ui/main.gd` saw Review above it. Moves lost their animation because assigning `board.game` ran the setter's `_end_slide()`. (2) Settings was still unreachable in a Review opened from a focus-mode game: `_set_focus` in `ui/main.gd` hides it and `ui/game_over.gd`'s Review never restores it. (3) after an agreed draw in focus mode, the non-modal draw notice counted as a covering screen and blocked Ctrl+, for up to 3 seconds (`ui/main.gd`).

REV-008: a two-row *engine lines* block sits between the eval graph and the move list. Each row has an eval chip (White's view) and the line in SAN, with the depth on the first row. Game positions show their stored lines. In a side line the Review starts its own Stockfish (MultiPV 2, unlimited search on the shown position) and pauses the background analysis queue until you leave the line. Clicking a move in a line plays up to it. Main files `engine/live_analysis.gd` (new), `ui/review.gd`, `engine/uci.gd`. Not reopened.

REV-009: each side-line move gets its class icon (on the square and in its row) once both its positions have settled: the analysis node budget, or converged depths. The same `Classification.evaluate` as game moves computes it, and it then stays fixed. The green "should have played" arrow works on side-line moves too. Main files `engine/live_analysis.gd`, `ui/review.gd`. Reopened once: (1) a game that ended in a draw has stored terminal analysis with an empty PV. Reused as the before-position of a legal explored move, it made `Classification.evaluate` read out of bounds and the move stayed unclassified (`ui/review.gd`, `_line_positions`). Returning to an already settled position also stopped the live search and showed the older frozen lines: `_next_fen` in `engine/live_analysis.gd` returned nothing once a class existed.

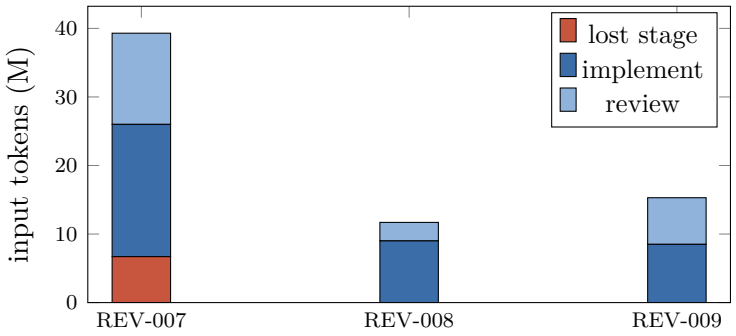
No roadmap image: P-Chess's `AGENTS.md` has no `Roadmap:` binding.

Timeline

17:22	Clone made from `184a0bf`, `godot --import`; the generated untracked `.uid`/`.import` files removed so Preflight sees a clean tree.
17:23–17:37	Launch 1, default lineup (Terra). REV-007 stage 2 codes without `ticket-flow`: no branch, no commit; parked as incomplete handoff, work stashed for recovery.
17:38	Foreman links the Agent-Skills skills into `~/.agents/skills` (the README's Codex path); `codex exec` lists `ticket-flow`. REV-007 back to `to-implement`.
17:39–17:49	Launch 2 (the one environmental restart). REV-007 to review, test-first (10 min).
17:55	Reopen 1: mouse input, Settings access, animation. Proxy widens `ui/main.gd` scope (17:56).
17:56–18:20	Repair (19 min); reopen 2: Settings after a focused game. Driver parks REV-007 and stops.
18:21	Foreman diagnosis committed; launch 3. REV-007 now on the hard model (Sol xhigh).
18:22–18:38	Repair (12 min); reopen 3: draw notice blocks Settings for 3 s.
18:38–18:52	Repair (10 min); merged (4 min).
18:52–19:22	REV-008 to review (25 min), merged (5 min).
19:23–19:46	REV-009 to review (18 min), reopened (5 min).
19:46–20:04	Repair (13 min), merged (5 min). Nothing left; PR #42.

Per stage

When	Ticket	Stage	Model	Min	In	Cache	Out	Cost	Outcome
17:37	REV-007	impl	terra xhigh	13	6.7M	99%	16k	0.49	incomplete handoff
17:49	REV-007	impl	terra xhigh	10	5.7M	99%	12k	0.41	to-review
17:55	REV-007	review	sol xhigh	6	4.5M	96%	22k	0.97	reopened
18:15	REV-007	impl	terra xhigh	19	6.9M	99%	17k	0.50	to-review
18:20	REV-007	review	sol xhigh	5	3.2M	95%	15k	0.79	reopened
18:33	REV-007	impl	sol xhigh	12	4.1M	98%	17k	0.72	to-review
18:38	REV-007	review	sol xhigh	5	3.3M	96%	14k	0.74	reopened
18:48	REV-007	impl	sol xhigh	10	2.6M	97%	12k	0.50	to-review
18:52	REV-007	review	sol xhigh	4	2.3M	94%	10k	0.60	merged
19:18	REV-008	impl	sol xhigh	25	9.0M	99%	50k	1.62	to-review
19:22	REV-008	review	sol xhigh	5	2.7M	94%	14k	0.70	merged
19:41	REV-009	impl	sol xhigh	18	5.3M	98%	36k	1.05	to-review
19:46	REV-009	review	sol xhigh	5	3.4M	94%	18k	0.88	reopened
19:59	REV-009	impl	sol xhigh	13	3.2M	98%	21k	0.67	to-review
20:04	REV-009	review	sol xhigh	5	3.4M	96%	17k	0.76	merged
				155	66.3M		291k	11.40	



Wasted: the first REV-007 stage (6.7M input, $0.49), whose work was stashed but whose handoff was unusable. REV-007 took 60% of the run's input (39.3M of 66.3M) and $5.73 of $11.40. Its last two reopen rounds ($1.53 review + $1.22 repair) fixed real but narrow variants of one criterion.

Comparison with previous runs

From `run-log.md`. The previous three-ticket lane, `qa-layout`: $5.78 for 3 ($1.93 each), one reopen, every handoff reaching review on the first attempt. `ui-screens`: $13.62 for 3 ($4.54 each), two reopens. This run: $11.40 for 3 ($3.80 each), four reopens, one lost stage. Two of the three tickets were `Difficulty: hard` and REV-007 escalated, so 10 of 15 stages ran on Sol xhigh, against none in `qa-layout`. Reviews cost $0.60–0.97 ($0.54–1.08 in `qa-layout`). REV-008's 25-minute implementation is the lane's longest stage.

Observations per model

- **gpt-6.1-terra xhigh, stage 2** (3 stages, REV-007 only): without the skill it wrote a plausible implementation (gate green, 526 tests) but no handoff at all. With the skill it worked test-first, with red/green commit pairs. Its tests drove the board through `board.play()` and `preferences.set_value()` instead of real input, which is how all three round-1 findings slipped. Its repair fixed the reported path but not the focused-game entry path.
- **gpt-6.1-sol xhigh, stage 2 (hard)** (5 stages): REV-008 went through on the first review, with red/green pairs for each behaviour (handshake, PV2, stop/error, queue). Its REV-007 repair for the draw notice special-cases that one node in `_covers_settings()` rather than restricting the

check to real screens; it passed, since the notice was the only offender. On REV-009 it misread criterion 7 ("searches nothing new for its class") as "no live search on return" and wrote tests asserting it.

- **gpt-6.1-sol xhigh, review** (7 stages, 4–6 min): every finding came with a real viewport or Xvfb reproduction and an external probe on the hard disk. It drip-fed REV-007, though: round 1 named only the `_covers_settings` path of criterion 8, and round 2 missed the draw-notice variant while claiming focused-entry coverage. It authored no fixes; it filed CLEAN-014 for long-line scrolling instead of adding it to REV-007.
- **opus-5.5 xhigh, proxy**: one call, granting REV-007's stage 2 the `ui/main.gd`/`ui/review.gd` scope for criteria 1, 2 and 8. All three later reviews checked the repairs stayed inside it.

Driver and runtime issues

1. **Codex had no `ticket-flow` on the Linux box.** `~/.agents/skills` did not exist (the earlier P-Chess Codex runs were on Windows), and `~/.codex/skills` holds only Codex's own skills. EasyMath worked anyway because its `AGENTS.md` says a bare-id prompt comes from the driver and must load `ticket-flow`, and Codex then searched `~/.claude/skills`. P-Chess's `AGENTS.md` has no such line. The driver parked the bad handoff correctly and stashed the work (`a3bea32`); its ping checks the proxy from each runtime, but not that the runtime can see `ticket-flow`.
2. **New clone needs an import and then a clean.** `godot --headless --import` writes 33 untracked `.uid`/`.import` files that `main` does not track yet (the same ones sit untracked in the main checkout); Preflight would refuse the dirty tree.
3. **Third reopen not parked.** The driver parks only at the second reopen. REV-007's third reopen went straight back to stage 2, which was the right call here (the cause was already diagnosed), but no rule covers a fourth.
4. No crash, timeout or API error. Shutdown 0.5–2.3 s per stage. Codex reached the proxy directly in all three launches.

Recommendations

1. Add EasyMath's line to P-Chess's `AGENTS.md` ("a bare-id prompt comes from the sweatshop driver; load `ticket-flow` and run the ticket's current stage"), so a missing skill install cannot repeat issue 1. The `~/.agents/skills` links are in place on this box now.
2. Extend the driver's per-runtime ping to confirm the runtime lists `ticket-flow` (as the foreman's `codex exec` check did), and stop before the first stage when it does not (issue 1).
3. Commit the 33 generated `.uid`/`.import` files to `main`, so new clones and the main checkout stay clean after an import (issue 2).
4. For UI tickets whose criteria name an interaction (drag, a shortcut, a Settings change), write in the ticket's test list that the test drives real viewport input. Every one of REV-007's round-1 findings came from tests that called the method behind the input.
5. Lineup: keep the default Terra lineup with the hard escalation. Scoreboard rows: `gpt-6.1-terra xhigh` → `gpt-6.1-sol xhigh` 101 reviews, 29% reopened (S2 ≥28%), and `gpt-6.1-sol xhigh` → `gpt-6.1-sol xhigh` 22 reviews, 23% (S2 23%). The Sol pair has more than 10 reviews, but its 22 are mostly hard tickets, so the two are not like for like.

Human debt and proxy decisions

No `Débito humano:` lines in this run's tickets. One proxy call, by `opus-5.5 xhigh` (Codex reached the proxy directly):

- REV-007 (stage 2, after reopen 1): `Proxy decided (opus-5.5 xhigh)`: "Yes. Stage 2 may widen both scopes, limited to the bullets below. Criteria 1, 2 and 8 can't be met without these changes, both files are already Primary files, and review R2 sent this question to the proxy.

Precedent: BOT-005 and CLOCK-001 widened scope the same way." (receipt `/tmp/sweatshop-proxy/20261105-175641-REV-007-2bed17.json`)

Acceptance

Checked against the spec's five acceptance scenarios on the integrated session. The code is identical to REV-009's gated head, so its gate is reused: 554 passed, 0 failed, with the real-Lc0 checks and one pixel check skipped. The graphical checks come from the stage-3 reviews under Xvfb, and the record is in `.scratch/review-explore/spec.md`, section Acceptance.

1. Explore a blunder with real Stockfish: **pending**, Bruno on Windows. Nothing ran a real engine.
2. Rematch from a side line: **pass with fakes** (`tests/test_review.gd`: `test_side_line_freezes_queued_game_progress_stop_works_and_rematch_keeps_pause`).
3. Unanalysed or queued game: **pass with fakes** (the same test and `test_unanalysed_side_line_settles_history_for_miss_and_prunes_replacements`).
4. Arrow setting: **pass**, graphical (Settings over Review, after a focused game, during the draw notice).
5. 800×600: **pass** on agent captures (`test_review_fits_an_800_by_600_window`, REV-008's reviewed live capture); Bruno's look pending.

Not examined: real engine output and timing, Threads = cores−1 and CPU load, physical promotion input, and manual play of each exit path.

Reopen attribution

Labelled by the foreman (opus-5.5); the reviewer was `gpt-6.1-sol xhigh`, so no self-judging. Rows appended to `.scratch/reopen-attribution.md`.

Ticket (round)	Finding	Label	Why
REV-007 (1)	Review board ignored real mouse input	S2-explícito/código	criterion 1 names drag and click-click
REV-007 (1)	Settings could not open over the Review	S2-explícito/código ?	criterion 8; borderline S1, the repair needed a proxy scope call
REV-007 (1)	Played moves lost their animation	S2-explícito/código	criterion 10 names animation length
REV-007 (2)	Settings unreachable after a focused game	S2-explícito/código	criterion 8, a supported entry path; review miss and drip-feed
REV-007 (3)	Draw notice blocked Settings for 3 s	S2-explícito/código ?	criterion 8; borderline implicit (P3, transient); review miss and drip-feed
REV-009 (1)	Stored terminal-draw analysis reused as a before-position	S2-implícito	usable inputs for criteria 2 and 4; the terminal shape is an existing convention
REV-009 (1)	Returning to a settled position stopped live analysis	S2-explícito/código ?	criterion 5 and REV-008's live search; borderline S1, criterion 7's wording invited it

Totals: S2-explícito/código 6 (3 borderline), S2-implícito 1; no S1, no S3. Fixed in review: none (the merging reviews authored no behaviour or proof). Review miss and drip-feed: REV-007 rounds 2 and 3, both on criterion 8. Review-induced churn: none. No session-review, release-audit or post-release findings yet.

Scoreboard

`scoreboard.ps1` over the ClassyCall, EasyMath, P-Chess, PhysiSyst, SIGAA-ME, SynchroNice and slip main checkouts (P-Chess with this run's rows, attribution and interventions merged). Moved

since the `qa-layout` report (EasyMath's lanes ran in between, and `slip` is new to the list): implementer `gpt-6.1-sol xhigh` 10 → 22 stage runs, still none failed; pair `gpt-6.1-sol xhigh` → `gpt-6.1-sol xhigh` 10 → 22 reviews, 10% → 23% reopened (this run added 3 of its 5 reopens); pair `gpt-6.1-terra xhigh` → `gpt-6.1-sol xhigh` 34 → 101 reviews, 38% → 29%. Interventions: automatic-recovery 114 → 130, proxy-decision (opus-5.5 xhigh) 19 → 47, human-rescue 32 → 33.

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| Implementer | Stage runs | To review | Failed | Asked | Median min to review | Cost |
|---|---|---|---|---|---|---|
| gpt-6.1-sol high | 311 | 275 | 12 | 24 | 23m | $839.52 (309/311 priced) |
| opus-5.5 xhigh | 24 | 16 | 8 | 0 | 14m | $100.17 (22/24 priced) |
| gpt-6.1-sol max | 41 | 29 | 9 | 3 | 24m | $84.42 (33/41 priced) |
| opus-5.5 high | 37 | 21 | 15 | 1 | 12m | $73.54 (29/37 priced) |
| gpt-6.1-sol xhigh | 22 | 22 | 0 | 0 | 18m | $57.42 (22/22 priced) |
| gpt-6.1-terra high | 59 | 30 | 27 | 2 | 9m | $9.45 (34/59 priced) |
| gpt-6.1-terra xhigh | 130 | 102 | 18 | 10 | 18m | $107.36 (130/130 priced) |
| gpt-6-astra medium | 57 | 47 | 1 | 9 | 10m | $162.72 (57/57 priced) |
(other implementer rows omitted: unchanged models outside this lineup)

| Implementer | Reviewer | Reviews | Reopened | Reopen rate | Reopened for S2 | S2 rate | Unclassified | Fixed in review for S2 |
|---|---|---|---|---|---|---|---|---|
| gpt-6.1-sol high | gpt-6.1-sol xhigh | 176 | 33 | 19% | 32 | 18% | 0 | 20 |
| gpt-6.1-sol max | gpt-6.1-sol xhigh | 9 | 2 | 22% | 2 | 22% | 0 | 0 |
| gpt-6.1-sol xhigh | gpt-6.1-sol xhigh | 22 | 5 | 23% | 5 | 23% | 0 | 1 |
| opus-5.5 high | gpt-6.1-sol xhigh | 1 | 1 | 100% | 1 | 100% | 0 | 0 |
| gpt-6.1-terra high | gpt-6.1-sol xhigh | 30 | 7 | 23% | 7 | 23% | 0 | 0 |
| ? | gpt-6.1-sol xhigh | 7 | 2 | 29% | 1 | 14% | 0 | 0 |
| gpt-6.1-terra xhigh | gpt-6.1-sol xhigh | 101 | 29 | 29% | 28 | >=28% | 1 | 12 |
(pairs under other reviewers omitted)

| Implementer -> Reviewer | Rounds | S2 explicit code | S2 explicit test | S2 implicit | S1 | S3 noise |
|---|---|---|---|---|---|---|
| gpt-6.1-sol xhigh -> gpt-6.1-sol xhigh | 5 | 6 | 0 | 2 | 0 | 0 |
| gpt-6.1-terra high -> gpt-6.1-sol xhigh | 9 | 10 | 1 | 3 | 1 | 0 |
| gpt-6.1-terra xhigh -> gpt-6.1-sol xhigh | 40 | 50 | 2 | 12 | 4 | 0 |

| Discovery | Recorded findings | Earlier review miss (yes / known) | Review-induced churn (yes / known) |
|---|---|---|---|
| release-audit | 2 | 2 / 2 | 0 / 2 |
| session-review | 36 | 29 / 31 | 0 / 31 |
| ticket-review | 395 | 9 / 256 | 1 / 258 |
Runtime completion to process exit: 1618.016s total over 1196 measured stages; 773 unmeasured.
| Intervention | Recorded events |
|---|---|
| automatic-recovery | 130 |
| human-rescue | 33 |
| proxy-decision (gpt-6-astra high, fallback) | 58 |
| proxy-decision (opus-5.5 xhigh) | 47 |
| proxy-decision (opus) | 42 |
| proxy-decision (supervisor) | 12 |
| proxy-decision (unknown) | 37 |
| routine-approval | 18 |
```

Under the same reviewer, Terra xhigh (101 reviews) reopens more than Sol xhigh (22) and Terra high (30). The Sol xhigh pair only sees hard or escalated tickets, so it is not a like-for-like comparison.

Pending for the human

1. On Windows with real Stockfish, before merging PR #42 (REV-008 and REV-009 are `Review: human`): open an analysed game from History at a Blunder, drag another move, watch depth rise and the icon arrive; click moves in an engine line; Back to game; Rematch from a side line; explore a queued game and watch the queue pause and resume. Watch CPU load.
2. By hand, post-game and from History: start a side line, overwrite it, walk it, leave it each way; toggle Best move arrow in Settings while the Review is open, including after an F11 game.
3. Decide on adding the "bare id loads `ticket-flow`" line to P-Chess's `AGENTS.md` (recommendation 1).
4. Triage CLEAN-014 (long side lines do not scroll to the current row).
5. After merging PR #42, tell the other open lanes' foremen to sync.